Structure & Function of Blood Vessels

Exam code: 0610 & 0970

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Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
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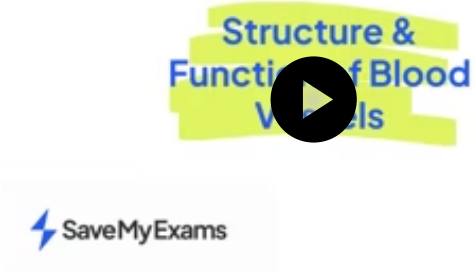

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Help me understand

How Structure of Blood Vessels is Adapted to their Function: Extended

Extended Tier Only

Arteries

- Have thick muscular walls containing elastic fibres **to withstand the high pressure of blood and maintain the blood pressure as it recoils after the blood has passed through**
- Have a narrow lumen **to maintain high pressure**

Veins

- Have a large lumen **as blood pressure is low**
- Contain valves **to prevent the backflow of blood as it is under low pressure**

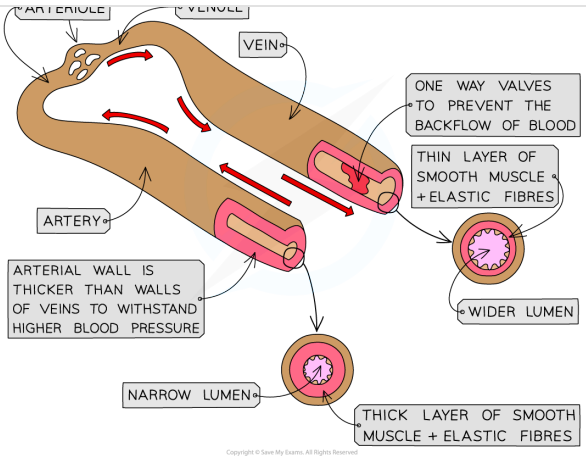
Capillaries

- Have walls that are one cell thick **so that substances can easily diffuse in and out of them**
- Have 'leaky' walls **so that blood plasma can leak out and form tissue fluid surrounding cells**

Stuck on something?

Ask questions and get help with exactly what you're revising.

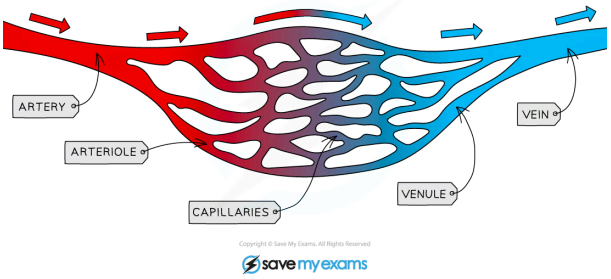
Got it



Adaptations of blood vessels

Arterioles and venules

- As arteries **divide more** as they get further away from the heart, they get **narrower**
- The narrow vessels that connect arteries to capillaries are called **arterioles**
- Veins also get narrower the further away they are from the heart
- The narrow vessels that connect capillaries to veins are called **venules**



The blood vessel network

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